

Project', was for the upliftment of the tribals of Jhabua, Madhya Pradesh. It provided them free education modules as well as key agriculture-related inputs.

In those days, Zee and Star TV used to uplink programmes from Hong Kong as the facilities were not available within India. As a team, we worked to implement digital satellite broadcasts for VSNL (now Tata Communications) whereby Star TV and Zee TV began their uplinks. Many such bandwidth-saving digital programmes began to be uplinked from India around this time, thus saving the huge costs of sending signals to Hong Kong and paying for foreign uplink stations.

Going forward, I made strategic tie-ups with various international companies like Scopus Israel, Scientific Atlanta USA (now Cisco), Azure Taiwan, etc, and brought in direct-to-home (DTH) technology to India. Those days, uplink in Ku-band frequency was banned by the government of India, so the team demonstrated the advantages of Ku-frequencies (used for DTH) to ISRO, got approval from the Ministry of I&B and went on to launch India's first DTH for the Andhra Pradesh government. Both uplinking and downlinking was set up and the then CM of the undivided state, Chandrababu Naidu, inaugurated this project, called APNET.

I then went on to find solutions for how to make the largest automatic antenna plant and tied-up with an international company, which had built the plants for Audi, Mitsubishi, Mercedes, etc. With the help of this firm, we set up a factory at Gandhinagar, Gujarat. This led to our company becoming the largest antennae manufacturing company in India.

In 2011, I started afresh with a different product line in the cable TV industry. I developed this division without any financial support, letter of credit, a bank guarantee or cash credit facilities. I initially struggled a lot as all the experienced staff were in the DTH division, and one of our most trusted and key marketing persons had left to join a competing Chinese company.

I strongly believe that the universe helps those who have faith in themselves. I got help from foreign equipment suppliers who trusted me. They provided products on very fair credit terms, and I was able to repay some of

them before the due dates.

I could then get a 90 per cent market share with the new product line, and this helped me grow at a rate of 200 to 400 per cent, year on year, for the next four years to reach a turnover of ₹ 580 million in 2015. MCBS then crossed a total turnover of one billion rupees (including existing DTH, the new electronics, STB and CATV divisions combined).

### Going the extra mile

A few more strategic tie-ups helped MCBS design the set-top box (STB) software as well as hardware, in Gandhinagar itself—something that had happened in India for the first time. We got these set-top boxes manufactured in other factories within India, and trials went on for over two years. This proved that these types of set-top boxes with complex electronics and high levels of security, similar to that used in ATM machines, could be designed and manufactured within India!

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In 2013, we began an intensive research project for developing technology to manufacture electronics in India. We began buying land and got the building started. Within eighteen months, the research was completed and, meanwhile, we convinced a chipset company to provide support to design STBs without committing to very large quantities. We started making STBs based on our own design in the factory of a Korean firm in Pune, even though it had old machines, just to gain confidence and get our newly recruited staff trained.

We placed an order for the world's most state-of-art high-speed robotic machines. The machines have robotic hands which can pick the minutest of electronic components that can't be seen clearly by human eyes.

The machines were commissioned in 2017 and set-top box production began soon thereafter. In 2019, the pro-

duction plant was upgraded to match the requirements laid down by ISRO to make the facility space worthy!

It was my dream to set up the largest electronics set-top box manufacturing plant within India with state-of-art machines, and this has been my major contribution to the company. I was discouraged by my own family members initially, as new investments could be a dent if markets crashed.

But the strategy to constantly but continuously invest year on year from whatever was earned in business helped me set up the plant, which required investments to the tune of ₹ 400 million. This made us India's largest single set-top box manufacturer to have grown from scratch.

Recently, in 2019, I have upgraded the electronics facility for manufacturing space and military grade equipment, and this has been approved by ISRO. Our plant can make electronics to be used in Indian satellites, moon missions, aerospace and missiles.

My company has earned many orders for technologies which were much ahead of their time like:

- Delivery of 'Video over Mobile' to Reliance Infocom way back in 2002-03.
- Broadcasting high-definition video during the Commonwealth Games 2010. This was the very first time HD video was used for any international games.
- Developing the commercial DTH market for broadcasting more channels per satellite transponder. We bagged tenders at Doordarshan for FreeDish, later expanding to cover Tata Sky and Videocon D2H. At this time, all antennae, etc, were being imported by the company.

Five years down the line, I want to devote my time to making new AI technology products, as artificial intelligence is going to take over the world sooner than expected. I want to make sure that I am a part of that revolution.

My advice to the new entrepreneurs is to begin a startup with some innovative technologies because the old or existing technologies are on the verge of being phased out. So, if they do the same thing that the others are doing, then there is not going to be any growth or appreciation. **EFY**

*Please read January 2020 issue of Electronics Bazaar for the full story*